

Seminar 5: Sampling and Aliasing

Topics: sampling continuous-time signals, normalized frequency, aliasing, folded aliases, the sampling theorem, spectral interpretation, oversampling and undersampling, stroboscopic aliasing, discrete-to-continuous reconstruction, and interpolation.

Key idea. Sampling replaces a continuous-time waveform by values taken at equally spaced instants. Different analogue frequencies can produce the same sequence, so exact reconstruction requires a sufficiently high sampling rate and an appropriate reconstruction method.

1. Sampling a continuous-time sinusoid

Consider the continuous-time sinusoid $x(t) = 4\cos(2\pi 300t + \pi/6)$.

- (a) Sample the signal at $f_s = 2000$ samples/s and write the resulting sequence $x[n]$.
- (b) Determine the sampling period T_s .
- (c) Determine the normalized cyclic frequency in cycles/sample and the normalized radian frequency in rad/sample.
- (d) How many samples are taken during one period of the sinusoid?
- (e) Repeat the normalized-frequency calculation for $f_s = 800$ samples/s. Is the sinusoid still sampled above the Nyquist rate?

2. Sampling a multi-sinusoidal signal

Consider $x(t) = 2 + 3\cos(2\pi 400t - \pi/4) + 2\sin(2\pi 900t)$.

- (a) Sketch the two-sided spectrum of $x(t)$, including the DC component, and label all frequencies and complex amplitudes.
- (b) Determine the highest frequency present in the signal.
- (c) Determine the theoretical minimum sampling rate required by the sampling theorem.
- (d) Would $f_s = 2000$ samples/s be sufficient? Would $f_s = 1600$ samples/s be sufficient? Explain.
- (e) Propose a practical sampling rate that provides some oversampling margin.

3. Different analogue frequencies, identical samples

A discrete-time sinusoid is $x[n] = \cos(0.4\pi n + \pi/5)$.

- (a) Assume $f_s = 1000$ samples/s. Find one continuous-time sinusoid with frequency between 0 and 500 Hz that produces $x[n]$.
- (b) Find two additional positive analogue frequencies below 2000 Hz that generate the same sample values.
- (c) Find one folded alias and determine the phase required so that its samples are identical to $x[n]$.
- (d) Explain why the original analogue frequency cannot be determined uniquely from $x[n]$ alone.

4. Families of aliases

A sinusoid $x(t) = A\cos(2\pi f_0 t + \phi)$ is sampled at f_s samples/s.

- (a) Write the family of analogue frequencies $f = f_0 + kf_s$ that produce the same normalized frequency.
- (b) Using cosine symmetry, write the family of folded aliases.

- (c) State what happens to the phase when a frequency is folded.
- (d) For $f_0 = 180$ Hz, $f_s = 500$ Hz, and $\phi = \pi/3$, list all positive aliases below 1200 Hz and give the phase associated with each.

5. Sampling theorem and oversampling

A measurement system records a signal known to contain no frequency components above 7.5 kHz.

- (a) According to the sampling theorem, what condition must the sampling rate satisfy for exact reconstruction?
- (b) Is a sampling rate of 15 kHz sufficient in the strict statement of the theorem? Explain.
- (c) Would 20 kHz be a valid choice? What is the Nyquist frequency in that case?
- (d) Explain why practical systems normally sample somewhat above the theoretical minimum.
- (e) If the signal is sampled at 48 kHz, calculate the oversampling factor relative to twice the highest signal frequency.

6. Discrete-time frequency and the principal Nyquist interval

Consider the sampled sinusoid $x[n] = \cos(1.6\pi n)$.

- (a) Reduce the normalized radian frequency to an equivalent value in the principal interval $-\pi \leq \omega_{\text{hat}} \leq \pi$.
- (b) Write $x[n]$ using that principal-interval frequency.
- (c) Sketch the spectral line locations over the normalized-frequency range $-3\pi \leq \omega_{\text{hat}} \leq 3\pi$, showing the 2π periodicity.
- (d) Explain why the discrete-time spectrum repeats every 2π .

7. Undersampling and frequency folding

A sinusoid of frequency $f_0 = 1000$ Hz is sampled at each of the following rates: 3000, 2200, 1800, 1250, and 800 samples/s.

- (a) For each sampling rate, determine whether the sampling theorem is satisfied.
- (b) For every undersampled case, determine the alias frequency that lies between 0 and $f_s/2$.
- (c) For each case, calculate the corresponding normalized radian frequency in the principal interval.
- (d) Make a plot of apparent reconstructed frequency versus sampling rate and identify where folding occurs.

8. Sampling followed by ideal reconstruction

Consider the system in which $x(t)$ is sampled at f_s samples/s and then passed through an ideal discrete-to-continuous reconstruction stage, producing $y(t)$. Suppose $x(t)$ is bandlimited to $|f| \leq B$.

- (a) State the condition on f_s that guarantees $y(t) = x(t)$.
- (b) For $B = 4$ kHz, determine the minimum theoretical sampling rate.
- (c) If $f_s = 6$ kHz, describe qualitatively what happens to frequency components between 3 and 4 kHz.
- (d) Explain why ideal reconstruction cannot undo aliasing that has already occurred during sampling.

9. Spectral interpretation of sampling

A continuous-time cosine of frequency 600 Hz is sampled at $f_s = 2000$ samples/s.

- (a) Determine its normalized cyclic and normalized radian frequencies.
- (b) Draw the two spectral lines inside the principal normalized-frequency interval.

- (c) Show the nearest repeated spectral lines outside the principal interval.
- (d) Convert the repeated normalized-frequency locations back to analogue-frequency aliases.
- (e) Explain which analogue-frequency representative ideal reconstruction selects.

10. Stroboscopic aliasing

A disk rotates clockwise at 750 revolutions/minute and is observed only when a strobe flashes.

- (a) If the strobe flashes at 750 flashes/minute, describe the apparent motion of the disk.
- (b) Repeat for 375 flashes/minute and explain why the disk can again appear stationary.
- (c) For 806 flashes/minute, determine the angular advance or retreat observed from one flash to the next.
- (d) Determine the apparent rotation rate in revolutions/minute for the 806 flashes/minute case.
- (e) Relate this visual effect to aliasing of a sampled sinusoid.

11. Reconstructing a discrete-time sinusoid

A digital-to-continuous converter operates at $f_s = 4000$ samples/s and receives $x[n] = 3\cos(0.7\pi n - \pi/4)$.

- (a) Determine the analogue frequency selected by ideal reconstruction inside the Nyquist interval.
- (b) Write the reconstructed continuous-time sinusoid.
- (c) List two other positive analogue frequencies that would produce the same sample sequence before reconstruction.
- (d) What is the largest analogue frequency that this converter can reconstruct without ambiguity?

12. Sample-and-hold and linear interpolation

A sequence has sample values $x[0] = 0$, $x[1] = 1$, $x[2] = 0.5$, $x[3] = -1$, $x[4] = 0$.

The sampling period is $T_s = 1$ ms.

- (a) Sketch the zero-order-hold reconstruction from $t = 0$ to 5 ms.
- (b) Sketch the linear interpolation over the same interval.
- (c) Compare the continuity and smoothness of the two reconstructions.
- (d) Explain why neither method is generally identical to ideal bandlimited interpolation.
- (e) State one practical reason why zero-order hold is commonly used in digital-to-analogue conversion.

13. Interpolation as a sum of shifted pulses

A reconstructed signal is formed from samples $x[n]$ using an interpolation pulse $p(t)$: $\hat{x}(t) = \sum_{n=-\infty}^{\infty} x[n]p(t - nT_s)$.

- (a) Explain the role of the shift nT_s and the multiplier $x[n]$.
- (b) Describe the pulse $p(t)$ for a zero-order hold.
- (c) Describe qualitatively how a triangular pulse produces linear interpolation.
- (d) What property must an ideal interpolation pulse have at the sample instants so that the reconstructed signal passes exactly through all sample values?
- (e) Explain why the pulse shape affects frequencies introduced or attenuated by the reconstruction stage.

14. Sampling a chirp

Consider a chirp whose instantaneous frequency increases linearly from 0 Hz at $t = 0$ to 3000 Hz at $t = 1$ s. The signal is sampled at $f_s = 4000$ samples/s.

- (a) Write the instantaneous frequency $f_i(t)$ of the continuous-time chirp.
- (b) Determine the time at which the chirp first reaches the Nyquist frequency.
- (c) For times after this instant, determine the apparent reconstructed frequency inside $0 \leq f \leq f_s/2$.
- (d) Sketch the true instantaneous frequency and the aliased apparent frequency versus time from 0 to 1 s.
- (e) Describe what the corresponding sampled chirp would sound or look like near and beyond the folding point.

15. Integrated computational investigation

Use Python with NumPy and Matplotlib to investigate sampling and reconstruction.

- (a) Generate a 100-Hz continuous-time cosine on a dense time grid and sample it at 1000, 250, 180, and 120 samples/s.
- (b) For each sampling rate, overlay the sample points on the original waveform and identify whether aliasing occurs.
- (c) For each case, calculate the normalized radian frequency and its principal-interval equivalent.
- (d) Plot the apparent reconstructed alias frequency as the input frequency is swept from 0 to $2f_s$ for a fixed f_s .
- (e) For one sample sequence, compare zero-order-hold and linear interpolation.
- (f) Summarize how the time-domain plots, normalized-frequency calculations, and reconstruction plots illustrate the sampling theorem.